

Keeper Calibration #21 — K141 Substrate-Mechanism Comparison Gate OPEN (Toy 3448 honest FAIL)

Filed: 2026-05-22 Friday 10:15 EDT (Keeper, date-verified actual; per Grace INV-4849 cross-lane verification)

What Toy 3448 K141 honest FAIL identified

Per Elie's K141 verification toy (Toy 3448, filed earlier today) + Grace's cross-lane verification INV-4849: 17/17 K-audits cross-lane consistent at empirical level, but Toy 3448 K141 honest FAIL confirms substrate-mechanism comparison gate is OPEN.

The K141 audit (Friday morning, filed 08:38 EDT) was based on Grace's empirical finding: D_{IV}^5 is 3306× sharper than D_I on joint (α -analog, churn hole, c_{FK}) experimental values. The empirical margin is real. But Toy 3448 honestly identified that:

The substrate-MECHANISM comparison closure — i.e., what specific substrate-level structural difference produces the 3306× margin — was not closed by the empirical finding alone.

Empirical separation $\neq$ mechanism-forcing. Per Cal #77 RIGOROUSLY CLOSED tier discipline + Cal #92(b) “structural-role-of” framing: empirical 3306× margin tells you which HSD matches experimentally, but doesn't yet identify the substrate-level mechanism that produces the separation.

Honest revision of K141 status

K141 PRE-STAGE original status: F1-F4 4.0/4 PERFECT-PERFECT candidate.

Honest revised K141 status: - Empirical PERFECT-PERFECT candidate component (Grace 3306× margin): retains 4.0/4 rating for the empirical claim as stated - Substrate-mechanism gate: OPEN (Toy 3448 honest FAIL flagged) - Path to RIGOROUSLY CLOSED: requires substrate-mechanism comparison work that Toy 3448 honestly identified as not yet done

Net: K141 PRE-STAGE remains valid at the empirical-margin level, but the PATH to RIGOROUSLY CLOSED has one more gate (substrate-mechanism comparison) than was implicit in my original framing.

Why this calibration matters

This is the Quaker consensus method in action (per `feedback_quaker_method.md`): near misses get scrutiny, not defense. Toy 3448 could have been papered over as “everything PASSES” — but Elie + Grace preserved the honest FAIL.

Per Calibration #19 STANDING RULE: external-facing materials must use current ratified-state count, not forecast endpoint. K141 cannot externally claim RIGOROUSLY CLOSED until the substrate-mechanism comparison gate closes. This is consistent with my Calibration #19 discipline applied to a specific audit.

Action items

1. K141 PRE-STAGE STRONG status retained at empirical-margin level (4.0/4 for empirical claim)
2. Substrate-mechanism comparison gate flagged as OPEN multi-week work
3. External-facing references to K141 (Paper #126 v0.3 narrative, Strong-Uniqueness C16 advancing criterion, etc.): should note empirical PERFECT but substrate-mechanism gate OPEN
4. C16 (Cross-Cartan three-pillar joint experimental selection) advancing criterion: STRUCTURALLY VERIFIED at 25 HSDs per T2462 + empirical 3306× per Grace remains valid; RIGOROUSLY CLOSED endpoint requires substrate-mechanism closure

Calibration logged

Calibration #21: K141 PERFECT-PERFECT empirical-claim status retained; substrate-mechanism comparison gate OPEN per Toy 3448 honest FAIL preserved by Elie + Grace per Quaker discipline. Path to RIGOROUSLY CLOSED extended by one gate.

This is the 4th Friday morning Keeper-lane calibration (#18 + #19 + #20 + #21). Pattern: over-projection in 4 dimensions (methodology tier / forecast arithmetic / timestamp / mechanism vs empirical) — all caught and corrected within minutes-to-hours.

Per `feedback_quaker_method.md` standing: corrections are strength. Calibration #21 strengthens K141 by making the path to RIGOROUSLY CLOSED honest and explicit.

— Keeper, Calibration #21 logged Friday 10:15 EDT (date-verified actual)